Hall Ticket Number:

MR-22

Code: B424CS4

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, July/August-2024

Database Management Systems

(Common to CSE, IT, CSB, CSM & CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Illustrate different types of Attributes.	1	2	1	1,2
1. b)	List different types of data models.	1	1	1	1,2
1. c)	Describe the properties of relation model.	1	2	2	1,2
1. d)	Explain relation algebra operators.	1	2	2	1,2
1. e)	Discuss lossless join decomposition.	1	2	3	1,2,3,12
1. f)	Demonstrate EXCEPT keyword in SQL.	1	3	3	1,2,3,12
1. g)	Define Transaction State.	1	1	4	1,2,3,12
1. h)	What is time stamp ordering protocol?	1	1	4	1,2,3
1. i)	Define Global depth and Local depth in hashing.	1	2	5	1,2
1. j)	Explain Tree based Indexing.	1	4	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the structure of a DBMS in detail.	5	2	1	1,2
2. b)	Summarize the major disadvantages of file-processing system.	5	2	1	1,2
	OR				
2. c)	Describe different types of Integrity Constraints.	5	2	1	1,2
2. d)	Design an ER diagram for a bank management system that includes entities such as bank, employees, branch, customer, loan and account. Specify the relationships between these entities and their attributes.	5	5	1	1,2
3. a)	Discuss Referential Integrity with example.	5	2	2	1,2,3,12
3. b)	Explain Domain Relational Calculus with example.	5	2	2	1,2,3,12
	OR				
3. c)	Write the given queries in SQL. products(Product-Name, Product-Id, Price) i. Sort the products from highest to lowest price. ii. Sort the products alphabetically by Product-Name. iii. Select all Product-Name, Product-Id with Price greater than 80.	5	4	2	1,2,3,12
3. d)	i. Develop Salesman Table (salesperson ID, name, city and commission). ii. Create a view to locate the salespeople in the city '_New York'.	5	4	2	1,2,3,12
4. a)	Write the given queries in SQL Sailors(sid: integer, sname: string, rating: integer, age: real); Boats(bid: integer, bname: string, color: string); Reserves(sid: integer, bid: integer, day: date). i. Find all information of sailors who have reserved boat number 101. ii. Find the names of sailors who have reserved a red boat, and green boat iii. Find the names of sailors who have reserved at least one boat.	5	4	3	1,2,3,12

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4. b)	Differentiate Assertions and Triggers with suitable example.	5	3	3	1,2,3,12
	OR				
4. c)	Illustrate First and Second Normal form with example.	5	3	3	1,2,3,12
4. d)	Discuss the problems caused by redundancy in Schema refinement.	5	2	3	1,2,3,12
5. a)	Outline ACID properties of transaction.	5	2	4	1,2,3
5. b)	Explain serializability concept with example.	5	2	4	1,2,3,12
	OR				
5. c)	Demonstrate Validation-Based protocols.	5	3	4	1,2,3
5. d)	Interpret the procedures used to recover data from a DBMS.	5	2	4	1,2,3
6. a)	Explain Primary and Secondary indexes with example.	5	2	5	1,2,3
6. b)	Illustrate the factors that support to evaluate Hashing and Indexing.	5	2	5	1,2,3
	OR				
6. c)	Discuss about B^+ tree in detail with example.	5	2	5	1,2,3
6. d)	Illustrate Indexed Sequential Access Methods.	5	2	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome